Project Problem Framing - Lucy Crowe

1. Problem

On university campuses, student engagement is hindered by information fragmentation. Campus events, club meetings, guest lectures, and social gatherings are spread across many disconnected channels, mainly departmental email blasts, university Instagram accounts, GroupMe chats, physical posters, and official web portals.

This creates two problems.

- Students miss relevant academic, networking, and social events because they lack the time to check many channels every day, or they find out about an event after it has already happened.
- Student organizations and event organizers see low attendance despite high marketing effort, because their announcements get buried in fast-moving chat feeds or ignored in long email newsletters.

Current solutions fall short. Traditional university event portals depend on manual, form-heavy data entry by club officers, which leads to low adoption and stale calendars. Social channels are chronological, noisy, and hard to search.

The problem is worth addressing because campus involvement affects student retention, well-being, and career development. Preliminary observations suggest that most campus organizations market events mainly through Instagram flyers or GroupMe messages, while few regularly update the official student activities portal because of administrative friction.

2. Primary User

Primary user: Undergraduate and graduate students looking for campus involvement opportunities that fit their interests and class schedules.

User goal: Open a single app and see an aggregated feed of campus events without checking multiple apps or feeds.

Context: On a phone or laptop between classes, during study breaks, or on weekend afternoons, looking for immediate or upcoming activities.

Core pain point: Information fragmentation and limited time. Students do not have the bandwidth to read every email blast, scroll Instagram, and sift through GroupMe feeds to learn what is happening today.

Current alternatives: Scanning GroupMe notifications, scrolling university Instagram accounts, skimming email newsletters, and relying on word of mouth.

Adoption driver: The product acts as an automated catch-all hub that pulls messy announcements from the channels students already follow into one filtered discovery feed.

3. Proposed Product Direction

The product is an AI-powered campus event aggregator and discovery hub. Students arrive at an app that is already populated with events from email, Instagram, and GroupMe, so they do not have to enter anything themselves.

1. Aggregation and ingestion. The system collects unstructured content from key campus sources: email blasts, Instagram captions and flyers, and GroupMe announcements.
2. AI processing. The AI reads the messy text and images, extracts structured event details (title, date, time, location, host, tags), and flags missing information.
3. User action. The student opens the app to a unified event feed and filters by category or by availability.
4. Outcome. The student sees a clean event card with a short summary, a location tag, and an indicator of whether the event fits their free time.

The central value is turning scattered campus announcements into one structured calendar by automating the extraction work.

4. AI-Fit Rationale

AI task: Unstructured data extraction, multi-format parsing (text and image), summarization, and tag classification.

Inputs: Raw content from email newsletters, social media captions, GroupMe messages, and flyer photos. Inputs often contain slang, messy formatting, and implicit details such as a time with no date.

Outputs: Structured data containing event title, date, start and end time, location, host organization, tags, and a one-sentence summary.

User interaction: The AI runs in the background and structures incoming content. Students browse clean event cards. Submitters and organizers see a preview card they can verify before an event is published.

Product value: Fixed rules and keyword filters cannot reliably interpret informal language or read text embedded in a flyer image. AI bridges the gap between casual human communication and structured database entries.

Alternatives: Manual entry forms and traditional rule-based parsers. Forms fail because of user friction, and rule-based parsers fail when phrasing varies. AI can interpret context, implicit dates, and visual flyer content.

Risks and Limitations

- The AI may extract an incorrect date, invent a location, or create duplicates when the same event is posted on more than one channel.
- Safeguards include a human review step for submitted events and a report button on event cards that flags questionable entries for manual correction.

5. Non-Goals and Prototype Boundaries

The first working prototype has the following boundaries.

1. Native mobile application. The prototype is a mobile-responsive web application built in Replit, with no App Store or Android build.
2. Ticketing and payments. No RSVP ticket purchasing, paid event processing, or payment integrations.
3. Social features. No user-to-user messaging, follower graphs, or public user profiles.
4. Calendar syncing. No two-way integration with Google Calendar or Apple Calendar.

6. Preparation for the First Working Slice

End-to-End Slice Description

The first working slice demonstrates the core pipeline: ingest a raw campus post, run AI extraction, and populate the unified feed.

User entry: A tester opens the app and views the pre-seeded aggregated feed, or pastes a raw text snippet from a campus post into an add-source field.

System processing: The system sends the unstructured text to a language model, extracts structured metadata, tags the event type, and saves it to the database.

User result: The newly parsed event appears as a formatted event card in the unified feed alongside the pre-loaded events.

Completion criteria: A raw announcement from an external channel is turned into a clean, filterable card visible in the live feed.

Three Test Cases for Evaluation

1. Informal chat message. A casual GroupMe style message with emojis and relative time references, to test whether the AI resolves dates such as tomorrow.
2. Multi-event email newsletter. A long departmental email listing several speakers and times, to test whether the AI separates discrete events or correctly extracts the primary one.
3. Incomplete social post. An Instagram caption missing key details such as the room or time, to test whether the AI marks missing fields or sets placeholders instead of guessing.

7. AI-Use Record

AI tools used: ChatGPT (OpenAI) and Claude (Anthropic).

Representative prompts:

- Help me articulate a startup problem statement centered on campus information fragmentation across Instagram, GroupMe, and email.
- How should I define prototype boundaries for an event aggregator app so I do not get stuck building complex web scrapers during a short milestone?
- Draft three edge-case test scenarios for parsing informal campus announcements using a language model.

AI contribution: Helped structure the assignment sections, define the primary user, and set realistic technical scope boundaries for the Replit prototype, specifically batch seeding instead of live scraping.

Rejected or revised suggestion: The AI initially suggested centering the app on a manual submission form where students paste links. I revised this because the core value of the product is a pre-aggregated hub that collects events from existing platforms, rather than placing the effort on the user.

Elements requiring additional testing: Accuracy of date parsing for relative terms such as this Friday versus next Friday, when processing archived or real-time social media posts.